Abstract Algebra - Homework 2

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Problem 1

Write $|G| = p^\alpha m$ with $p \nmid m$ and $|P| = p^\alpha$. Because $N \trianglelefteq G$, Lagrange gives $|N| = p^\beta d$ where $0 \le \beta \le \alpha$ and $p \nmid d$. First consider $N \cap P$. Since $N \cap P \le P$, we have

$$|N \cap P| \mid |P| = p^\alpha \quad \Longrightarrow \quad |N \cap P| = p^\gamma$$

for some $0 \le \gamma \le \beta$; thus $N \cap P$ is a p-subgroup of N.

Next form $NP \le G$. Using $|NP| = \dfrac{|N||P|}{|N \cap P|}$ we compute

$$|NP| = \frac{p^\beta d\, p^\alpha}{p^\gamma} = p^{\alpha+\beta-\gamma} d, \qquad [NP : P] = \frac{|NP|}{|P|} = p^{\beta-\gamma} d.$$

Because $P \le NP \le G$, the index $[NP : P]$ divides $[G : P] = m$; since $p \nmid m$ we must have $p^{\beta-\gamma} = 1$. Hence $\beta = \gamma$ and $|N \cap P| = p^\beta$. This shows $N \cap P$ attains the maximal power of p dividing $|N|$, so $N \cap P$ is a Sylow p-subgroup of N.

For the quotient G/N we have

$$|G/N| = \frac{|G|}{|N|} = \frac{p^\alpha m}{p^\beta d} = p^{\alpha-\beta} \frac{m}{d},$$

so any Sylow p-subgroup of G/N has order $p^{\alpha-\beta}$. Since $N \le NP$, the subgroup $NP/N \le G/N$ satisfies

$$|NP/N| = \frac{|NP|}{|N|} = \frac{p^\alpha d}{p^\beta d} = p^{\alpha-\beta},$$

hence NP/N is a Sylow p-subgroup of G/N.

Thus $N \cap P$ is Sylow-p in N, and NP/N is Sylow-p in G/N.

Problem 2

(1) Naturally, since both P and Q are Sylow p-subgroups of the finite group G, the well-known Sylow theorem assures us P and Q are conjugate.

(2) **($\subseteq$)** Let $g \in N_G(P)$, so by definition $gPg^{-1} = P$. Since f is a homomorphism, we have:

$$f(g)f(P)f(g)^{-1} = f(gPg^{-1}) = f(P),$$

which implies $f(g) \in N_H(f(P))$. Therefore, $f(N_G(P)) \subseteq N_H(f(P))$.

($\supseteq$) Now let $h \in N_H(f(P))$. Since f is surjective, there exists $g \in G$ such that $f(g) = h$. Consider the conjugate subgroup $gPg^{-1} \leq G$. Then:

$$f(gPg^{-1}) = f(g)f(P)f(g)^{-1} = hf(P)h^{-1} = f(P),$$

so $f(gPg^{-1}) = f(P)$. By previous section, it follows that gPg^{-1} and P are conjugate in G; that is, there exists $x \in G$ such that $xgPg^{-1}x^{-1} = P$. Define $y = xg$, then:

$$yPy^{-1} = xgPg^{-1}x^{-1} = P \Rightarrow y \in N_G(P),$$

and since $f(y) = f(xg) = f(x)f(g)$, we get:

$$f(g) = f(x)^{-1}f(y) \in f(N_G(P)).$$

Thus $h = f(g) \in f(N_G(P))$, so $N_H(f(P)) \subseteq f(N_G(P))$.

(3) Since $P \leq G$ and f is a group homomorphism, we have $f(P) \leq H$, so $|f(P)| \mid |H|$. By the First Isomorphism Theorem, we have:

$$P/\ker(f|_P) \cong f(P),$$

so

$$|f(P)| = \frac{|P|}{|\ker(f|_P)|} = \frac{p^\alpha}{|\ker(f|_P)|},$$

which implies that $|f(P)| = p^\beta$ for some $\beta \in [0, \alpha]$.

Since $p \nmid |H|$ and $|f(P)| \mid |H|$, the only possibility is $|f(P)| = 1$. Therefore,

$$f(P) = \{e\} \subseteq \ker(f).$$

From previous section, we know that $f(N_G(P)) = N_H(f(P))$. Since $f(P)$ is trivial, its normaliser is all of H, so:

$$N_H(f(P)) = H \Rightarrow f(N_G(P)) = H.$$

Problem 3

(1) (a) By Sylow's theorems, the number n_5 of 5-Sylow subgroups satisfies $n_5 \equiv 1 \mod 5$ and $n_5 \mid \frac{|G|}{p^k} = \frac{15}{5} = 3$, so $n_5 = 1$. A unique Sylow subgroup is always normal, so G has a normal subgroup of order 5.

(b) Let P_3 and P_5 be the Sylow 3- and 5-subgroups of G, with $|P_3| = 3$ and $|P_5| = 5$. Since $|P_3| = 3$ and $|P_5| = 5$ are relatively prime, and $|P_3||P_5| = 15 = |G|$, it follows that $P_3 \cap P_5 = \{e\}$. Moreover, P_5 is normal in G by part (a), and both P_3 and P_5 are cyclic because all groups of prime order are cyclic. Therefore, the internal direct product P_3P_5 is a subgroup of G of order 15. Since its order equals $|G|$, we have:

$$G = P_3P_5 \cong P_3 \times P_5 \cong \mathbb{Z}/3 \times \mathbb{Z}/5 \cong \mathbb{Z}/15,$$

so G is cyclic.

(2) Again by Sylow's theorems, the number n_p of p-Sylow subgroups satisfies $n_p \equiv 1 \mod p$ and divides q, so $n_p = 1$ or q. Similarly, $n_q \equiv 1 \mod q$ and divides p, so $n_q = 1$ or p. Suppose for contradiction that $n_q = p$. Then:

$$p \equiv 1 \mod q \Rightarrow p \equiv 1 \pmod{q},$$

which contradicts the assumption $p \not\equiv 1 \pmod{q}$. Therefore, $n_q = 1$, and the q-Sylow subgroup is unique and hence normal.

All groups of prime order are cyclic, so both the p-Sylow and q-Sylow subgroups are cyclic. Let P and Q denote the p- and q-Sylow subgroups, with $Q \trianglelefteq G$ and $P \cap Q = \{e\}$. Since $|P||Q| = pq = |G|$ and Q is normal, we have:

$$G \cong P \times Q \cong \mathbb{Z}/p \times \mathbb{Z}/q \cong \mathbb{Z}/pq,$$

because p and q are relatively prime. Hence, G is cyclic.

(3) (a) Once again by Sylow's theorems, the number n_3 of 3-Sylow subgroups satisfies $n_3 \equiv 1 \mod 3$ and divides 4. Thus, $n_3 = 1$ or 4. The number n_2 of 2-Sylow subgroups satisfies $n_2 \equiv 1 \mod 2$ and divides 3. So $n_2 = 1$ or 3. If either $n_3 = 1$ or $n_2 = 1$, the corresponding Sylow subgroup is unique and hence normal. If both were non-unique (i.e.,

$n_3 = 4$ and $n_2 = 3$), then G would contain more than 12 elements, which is impossible. Hence, at least one of the Sylow subgroups is normal and therefore, G always has a normal subgroup.

(b) A non-cyclic group of order 12 is the alternating group A_4, the group of even permutations on 4 elements. It has order 12 but is not cyclic. A cyclic group of order 12 must have an element of order 12. However, in A_4, all elements have order 1, 2, or 3. Therefore, A_4 cannot be cyclic.

Problem 4

First, note that both a and b are products of two disjoint 4-cycles. Therefore,

$$|a| = |b| = \mathrm{lcm}(4, 4) = 4.$$

Since their cycles overlap, they do not act on disjoint sets, so we compute explicitly:

$$ab = (1234)(5678)(1537)(2648), \quad ba = (1537)(2648)(1234)(5678).$$

We compute

$$ab = (17)(28)(36)(45) = ba,$$

so a and b commute and H is abelian.

To determine the order of H, observe that a has order 4, and b also has order 4. We compute a^2, b^2, and their combinations to find the number of distinct elements generated. It turns out that:

$$H = \langle a, b \rangle = \{e, a, a^2, a^3, b, ab, a^2b, a^3b\}$$

which contains 8 distinct elements. So $|H| = 8$.

Since H is a finite abelian group of order 8, by the classification theorem for finite abelian groups, it must be isomorphic to one of:

$$\mathbb{Z}_8, \quad \mathbb{Z}_4 \times \mathbb{Z}_2, \quad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

We note that H contains elements of order 4 (e.g., a), but no element of order 8, so it cannot be cyclic. It also contains an element of order 4 and another of order 2 (e.g., b^2), so it is not the elementary abelian group $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, which has all elements of order 2.

Therefore, the group must be:

$$H \cong \mathbb{Z}_4 \times \mathbb{Z}_2.$$

Problem 5

(1) Assume G/Z is cyclic, so there exists $g \in G$ such that every element of G/Z is of the form $g^n Z$ for some $n \in \mathbb{Z}$. Then for any $x \in G$, there exists $n \in \mathbb{Z}$ and $z \in Z$ such that:

$$x = g^n z.$$

Now let $x = g^n z$ and $y = g^m z'$, with $z, z' \in Z$. Then:

$$xy = g^n z \cdot g^m z' = g^{n+m} zz', \quad \text{and} \quad yx = g^m z' \cdot g^n z = g^{m+n} z'z.$$

Since $Z \subseteq Z(G)$, we have $zz' = z'z$, so:

$$xy = yx.$$

Thus, any two elements of G commute, so G is abelian.

(2) Since G is a p-group, its center $Z(G)$ is nontrivial. Hence, $|Z(G)| = p$ or p^2. If $|Z(G)| = p^2$, then $Z(G) = G$, so G is abelian. If $|Z(G)| = p$, then the quotient group $G/Z(G)$ has order p and is cyclic. By previous section, if $G/Z(G)$ is cyclic and $Z(G)$ is central, then G is abelian. Therefore, in both cases, G is abelian.

(3) Let p be a prime. Up to isomorphism, there are exactly two groups of order p^2:

$$\mathbb{Z}_{p^2}, \quad \mathbb{Z}_p \times \mathbb{Z}_p.$$

These are the only abelian groups of order p^2, and by previous sectio, all groups of this order are abelian. Therefore, this list is complete up to isomorphism.

Problem 6

(1) Since $A \cong \mathbb{Z}_n$, any homomorphism $\chi \in \widehat{A}$ is determined by the image of a generator, and this image must be an n-th root of unity. Therefore, $\widehat{A} \cong \mu_n$, the group of n-th roots of unity, which is cyclic of order n. Hence, $\widehat{A}$ is cyclic of order n.

(2) By the structure theorem for finite abelian groups, A is isomorphic to a product of cyclic groups:

$$A \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}.$$

Then,

$$\widehat{A} \cong \widehat{\mathbb{Z}_{n_1}} \times \cdots \times \widehat{\mathbb{Z}_{n_k}},$$

and each $\widehat{\mathbb{Z}_{n_i}} \cong \mathbb{Z}_{n_i}$. Therefore, $\widehat{A} \cong A$, so $|\widehat{A}| = |A| = n$.